

Introduction to Contract Theory and Auctions

Bayesian Games

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- The Buyer-Seller example from one of the previous lecture was an example of a **Bayesian Game**.
- In these games there is uncertainty over the payoff relevant (or possibly payoff irrelevant also) information in given pure strategy profiles.

Definition 2.1

A Bayesian game is a tuple $G = (N, \{A_i\}_{i \in N}, \Omega, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N}, \{\mu_i\}_{i \in N}, \{u_i\}_{i \in N})$ where

- $(N, \Omega, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N}, \{\mu_i\}_{i \in N})$ is a belief space.
- A_i is the set of **actions** of player i . $A = \prod_{i=1}^n A_i$ is the set of **action profiles**.
- $u_i : A \times \Omega \rightarrow \mathbb{R}$ is players i 's (state-dependent) utility function.

Definition 2.2

A **pure strategy** for player i is a function $s_i : T_i \rightarrow A_i$. A strategy of a player is a choice of action given her private signal realization. The **set of pure strategies for player i** is S_i and is the collection of all functions from T_i to A_i . The set of **pure strategy profiles** is $S = \prod_{i=1}^n S_i$.

Definition 2.3

A **behavioral strategy** for player i is a function $\sigma_i : T_i \rightarrow \Delta(A_i)$, that is, it is a choice of a probability distribution over actions given her private signal realization. The **set of behavioral strategies for player i** , $\hat{S}_i$, is the collection of all functions from T_i to $\Delta(A_i)$. The set of behavioral strategy profiles is $\hat{S} = \prod_{i=1}^n \hat{S}_i$.

Example:

$$T_i = \{a, b\}, \quad A_i = \{0, 1, 2\}; \quad \Delta(A_i) = \{(p_0, p_1, p_2) : p_i \geq 0, p_0 + p_1 + p_2 = 1\}$$

$$= \{(p_0, p_1, 1 - p_0 - p_1) : 0 \leq p_0 \leq 1, 0 \leq p_1 \leq 1 - p_0\}$$

$$\sigma_i : T_i \rightarrow \Delta(A_i)$$

$$\sigma_i(a) = (p_0, p_1, 1 - p_0 - p_1)$$

$$\sigma_i(b) = (q_0, q_1, 1 - q_0 - q_1)$$

$$\sigma_1(a) = (1/2, 1/2, 0)$$

$$\sigma_1(b) = (1/3, 1/3, 1/3)$$

$$\sigma_2(a) = (1/3, 0, 2/3)$$

$$\sigma_2(b) = (1/4, 3/4, 0)$$

when types (a, a)

	0	1	2
0	$1/2 \cdot 1/3$	$1/2 \cdot 0$	$1/2 \cdot 2/3$
1	$1/2 \cdot 1/3$	$1/2 \cdot 0$	$1/2 \cdot 2/3$
2	$0 \cdot 1/3$	$0 \cdot 0$	$0 \cdot 2/3$

(a, b)

	0	1	2
0	$\frac{1}{2} \cdot \frac{1}{4}$	$\frac{1}{2} \cdot \frac{3}{4}$	$\frac{1}{2} \cdot 0$
1	$1/2 \cdot 1/4$	$1/2 \cdot 3/4$	$1/2 \cdot 0$
2	$0 \cdot 1/4$	$0 \cdot 3/4$	$0 \cdot 0$

Definition 2.4

Given a pure strategy profile $s = (s_1, \dots, s_n) \in S$, **player i 's ex-ante expected utility or payoff** is

$$E_{\mu_i, s}[u_i] = \int_{\Omega} u_i(s_1(\tau_1(\omega)), \dots, s_n(\tau_n(\omega)), \omega) d\mu_i(\omega)$$

Definition 2.5

Given a pure strategy profile $s = (s_1, \dots, s_n) \in S$, **player i 's interim expected utility or payoff** is

$$E_{\mu_i, s}[u_i | \tau_i = t_i] = \int_{\Omega} u_i(s_1(\tau_1(\omega)), \dots, s_i(t_i), \dots, s_n(\tau_n(\omega)), \omega) d\mu_i(\omega | \tau_i = t_i)$$

Definition 2.6

A **Bayes-Nash Equilibrium at the ex-ante stage** is a strategy profile $s^* = (s_1^*, \dots, s_n^*)$ such that no player can improve her ex-ante expected utility by changing unilaterally her strategy. That is, for all $i \in N$, all $s'_i \in S_i$, it holds that

$$E_{\mu_i, s^*}[u_i] \geq E_{\mu_i, (s'_i, s_{-i}^*)}[u_i]$$

Definition 2.7

A **Bayes-Nash Equilibrium at the interim stage** is a strategy profile s^* in which, for each player i and each type $t_i \in T_i$, it holds that

$$E_{\mu_i, s^*}[u_i | \tau_i = t_i] \geq E_{\mu_i, (s'_i, s_{-i}^*)}[u_i | \tau_i = t_i]$$

for all $s'_i \in S_i$.

Proposition 2.8

$BNE\text{-interim} \subseteq BNE\text{-ex-ante}$

Proof: s^* is BNE-interim

$$E_{\mu_i, s^*}[u_i] = E_{\tau_i}[E_{\mu_i, s^*}[u_i | \tau_i]] \geq E_{\tau_i}[E_{\mu_i, s^*, s^*}[u_i | \tau_i]] \\ = E_{\mu_i, s^*, s^*}[u_i] \text{ which holds for all } i, s_i \in S_i$$

$\therefore s^*$ is also a BNE ex-ante equilibrium.

Remark 2.9

Conversely, if Ω is finite and there is a common prior $\mu = \mu_1 = \dots = \mu_n$, then if s is an ex-ante BNE, then there is an $\tilde{s}$ that is an interim-BNE, and such that the probability, under μ , that $s_i(t_i) \neq \tilde{s}_i(t_i)$ is zero.

Example: An opponent of unknown strength

The row player's type is known, but the column player can be strong, with probability α , or weak, with probability $1 - \alpha$. The payoff matrices are:

		Column $\mathbb{I}$	
		F	Y
Row $\mathbb{I}$	F	-1, 1	1, 0
	Y	0, 1	0, 0

α

		Column $\mathbb{II}$	
		F	Y
Row $\mathbb{I}$	F	1, -1	1, 0
	Y	0, 1	0, 0

$1 - \alpha$

$$\Omega = \{W, S\}$$

$$\tau_R(W) = \tau_R(S) = t$$

$$\tau_C(W) = W$$

$$\tau_C(S) = S$$

$$\mathcal{P}_R = \{W, S\}$$

$$\mathcal{P}_C = \{W, S\}$$

$$\mu(W) = 1 - \alpha$$

$$\mu(S) = \alpha$$

$$\mu_C(W|W) = 1, \mu_C(W|S) = 0$$

$$\mu_C(S|S) = 1, \mu_C(S|W) = 0$$

$$\mu_R(W|\{W, S\}) = 1 - \alpha$$

$$\mu_R(S|\{W, S\}) = \alpha$$

$$A_i = \{F, Y\} \quad i \in C, R \quad \left| \quad \begin{array}{l} S_R = A_R = \{F, Y\} \\ S_C: \{W, S\} \rightarrow \{F, Y\} \\ S_C = \left\{ \begin{pmatrix} F \\ F \end{pmatrix}, \begin{pmatrix} F \\ Y \end{pmatrix}, \begin{pmatrix} Y \\ F \end{pmatrix}, \begin{pmatrix} Y \\ Y \end{pmatrix} \right\} \end{array} \right. \quad S_C = (\text{action } W, \text{ action } S)$$

When Strong, Y is a dominated strategy $S_C^*(S) = F$

If Row Fights she get :

-1	if	C Strong
1	if	C Weak

$$E_{\text{Row}}(\text{Fighting}) : (-1)\alpha + (1)\alpha = 1 - 2\alpha$$

If Row Yields, she gets : 0

Row : F if : $1 - 2\alpha > 0 \Leftrightarrow \frac{1}{2} > \alpha$
 Y if : $1 - 2\alpha < 0 \Leftrightarrow \frac{1}{2} < \alpha$

$$S_R^* = \begin{cases} F & \text{if } \alpha < 1/2 \\ Y & \text{if } \alpha > 1/2 \end{cases}$$

$$S_C^*(W) = \begin{cases} Y & \text{if } \alpha < 1/2 \\ F & \text{if } \alpha > 1/2 \end{cases}$$

$$BNE = \left\langle F, \begin{pmatrix} Y \\ F \end{pmatrix} \right\rangle$$

$$\alpha < 1/2$$

$$\alpha > 1/2 \left\langle Y, \begin{pmatrix} F \\ F \end{pmatrix} \right\rangle$$

Example: Classical Adverse Selection

A firm A is taking over another firm B . The true value of B is known to B but unknown to A ; it is only known to A to be uniformly distributed on $[0, 1]$. It is also known that B 's value will flourish under A 's ownership: it will rise to λx , where x is the pre-takeover value and $\lambda > 1$. This description is common knowledge.

$$\Omega = [0, 1], \quad N = \{A, B\}$$

$$T_A = \{t_i\} \quad T_A(x) = t_i \text{ for all } x \in [0, 1]$$

$$T_B = [0, 1] \quad T_B(x) = x$$

$$\mu \sim \text{Uniform } [0, 1] \sim 1 \sim \text{cdf } x$$

A strategy for firm A is a bid $y \in \mathbb{R}_+$

A strategy for firm B $(x, y) \mapsto S(x, y)$
value $\nearrow$ bid $\nwarrow$

A strategy for B is a function
 $(x, y) \mapsto b$
value $\nearrow$ bid $\nwarrow$

$$b^*(x, y) = \begin{cases} 1 & \text{if } y \geq x \\ 0 & \text{if } y < x \end{cases}$$

$$\text{Payoff of } A = \begin{cases} \lambda x - y & b(x, y) = 1 \\ 0 & b(x, y) = 0 \end{cases}$$

$$\begin{aligned}
 E[\text{Payoff of } A \mid \text{Acceptance}] &= E[\text{Payoff of } A \mid y \geq X] \\
 &= E[\lambda X - y \mid y \geq X] \\
 &= \lambda \underbrace{E[X \mid y \geq X]} - y = \lambda \frac{y}{2} - y = \underline{y(\frac{\lambda}{2} - 1)}
 \end{aligned}$$

$$E[X \mid y \geq X] = \frac{1}{c} \int_0^y x \, dx = \frac{1}{y} \int_0^y x \, dx = \frac{1}{y} \cdot \frac{y^2}{2} = \frac{y}{2}$$

$$c = \Pr(y \geq X) = \Pr(X \leq y) = y$$

$$E[\text{Payoff of } A \mid \text{Rejection}] = 0$$

$$\begin{aligned}
 E[\text{Payoff of } A \mid \text{Acceptance}] > 0 &\Leftrightarrow \frac{\lambda}{2} - 1 > 0 \\
 &\Leftrightarrow \lambda > 2
 \end{aligned}$$

Example: Information can hurt!

In contrast with the single-agent case, more information can hurt both the informed and uninformed parties. Consider the following scenario: there are two states and two corresponding payoff matrices:

		Column		
		L	M	R
Row	T	1, 2	1, 0	1, 3
	B	4, 4	0, 0	0, 5

$\omega_1 : \text{prob. } 1/2$

		Column		
		L	M	R
Row	T	1, 2	1, 3	1, 0
	B	4, 4	0, 5	0, 0

$\omega_2 : \text{prob. } 1/2$

$$N = \{R, C\},$$

$$\Omega = \{\omega_1, \omega_2\}$$

$$T_i = \{a\} \quad i=1,2$$

$$\tau_i(\omega_1) = \tau_i(\omega_2) = a, \quad i=1,2.$$

$$P_1 = \{\omega_1, \omega_2\}$$

$$P_2 = \{\omega_1, \omega_2\}$$

$$\mu(\omega_1) = \mu(\omega_2) = 1/2$$

$$S_R(a) \in \{T, B\}$$

$$S_C(a) \in \{L, M, R\}$$

Strategies are equivalent to actions

ponding payoff matrices:

		Column		
		L	M	R
Row	T	1, 2	1, 0	1, 3
	B	4, 4	0, 0	0, 5

$\omega_1 : \text{prob. } 1/2$

		Column		
		L	M	R
Row	T	1, 2	1, 3	1, 0
	B	4, 4	0, 5	0, 0

$\omega_2 : \text{prob. } 1/2$

"Combined matrix": Computing expected values for each strategy profile (actions)

		Column		
		L	M	R
Row	T	1, 2 ←	<u>1</u> , 1.5	<u>1</u> , 1.5
	B	4, 4 →	0, 2.5	0, 2.5

BNE is $\langle B, L \rangle$

Example: Information can hurt! (continued)

Now suppose that player 2 is informed, while player 1 is uninformed, as before.

		Column		
		L	M	R
Row	T	1, 2	1, 0	1, 3
	B	4, 4	0, 0	0, 5

		Column		
		L	M	R
Row	T	1, 2	1, 3	1, 0
	B	4, 4	0, 5	0, 0

$$T_R = \{a_i\}$$

$$T_c = \{\omega_1, \omega_2\}$$

$$\omega_1 : \text{prob. } 1/2$$

$$\tau_R(\omega) = \text{constant}$$

$$\tau_c(\omega_1) = \omega_1$$

$$\tau_c(\omega_2) = \omega_2$$

$$s_R \leftrightarrow \text{action}$$

$$(s_c(\omega_1), s_c(\omega_2))$$

$$\omega_2 : \text{prob. } 1/2$$

$$BR_c^*(T, \omega_1) = R \quad \text{and} \quad S_c^*(\omega_1) = R$$

$$BR_c^*(B, \omega_1) = R$$

$$BR_c^*(T, \omega_2) = M \quad \text{and} \quad S_c^*(\omega_2) = M$$

$$BR_c^*(B, \omega_2) = M$$

ponding payoff matrices:

		Column		
		L	M	R
Row	T	1, 2	1, 0	1, 3
	B	4, 4	0, 0	0, 5

$\omega_1 : \text{prob. } 1/2$

		Column		
		L	M	R
Row	T	1, 2	1, 3	1, 0
	B	4, 4	0, 5	0, 0

$\omega_2 : \text{prob. } 1/2$

$$E[T \mid R \text{ when } \omega_1, M \text{ when } \omega_2] = 1 \quad S_n^* = T$$

$$E[B \mid R \text{ when } \omega_1, M \text{ when } \omega_2] = 0$$

$$BNE = \langle T, (R, M) \rangle$$

Example: Infection

Suppose that $T_1 = \{a, b\}$, $T_2 = \{a', b'\}$ Suppose the type maps are given by

$$\tau_1(\omega_1) = a, \tau_1(\omega_2) = \tau_1(\omega_3) = b, \quad \text{and} \quad \tau_2(\omega_1) = \tau_2(\omega_2) = a', \tau_2(\omega_3) = b'$$

$$\mathcal{H} = \{\Theta_1, \Theta_2\}$$

$$\omega_1 \leftrightarrow \Theta_1$$

$$\omega_2, \omega_3 \leftrightarrow \Theta_2$$

		Column	
		L	R
Row	U	2, 2	0, 0
	D	3, 0	1, 1

ω_1 : prob. 9/13

Θ_1

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_2 : prob. 3/13

Θ_2

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_3 : prob. 1/13

$$\mathcal{P}_1 = \{\omega_1, \{\omega_2, \omega_3\}\}$$

$$\mathcal{P}_2 = \{\{\omega_1, \omega_2\}, \{\omega_3\}\}$$

Event: "The game is Θ_1 ": $\{\omega_1\}$

Event "the game is Θ_2 ": $\{\omega_2, \omega_3\}$

• At ω_1 $K_1 \{\omega_3\} = \{\omega_3\}$

$$K_2 \{\omega_1\} = \emptyset \Rightarrow (K_2 \{\omega_1\})^c = \Omega$$

$$K_1 [(K_2 \{\omega_1\})^c] = \Omega$$

$$\mathcal{P}_1 = \{\omega_1, \{\omega_2, \omega_3\}\}$$

$$\mathcal{P}_2 = \{\{\omega_1, \omega_2\}, \{\omega_3\}\}$$

Event: "The game is Θ_1 ": $\{\omega_1\}$

Event "the game is Θ_2 ": $\{\omega_2, \omega_3\}$

• At ω_1 $K_1 \{\omega_1\} = \{\omega_1\}$
 $K_2 \{\omega_1\} = \emptyset \Rightarrow (K_2 \{\omega_1\})^c = \Omega$

$$K_1 [(K_2 \{\omega_1\})^c] = \Omega$$

At ω_2

$$K_1 \{\omega_2, \omega_3\} = \{\omega_2, \omega_3\}$$

$\downarrow$

$$\Theta_2 = \{\omega_2, \omega_3\}$$

$K_2 \{\omega_2, \omega_3\} = \{\omega_3\}$: At ω_2 P2 still doesn't know the game.

$$(K_2 \{\omega_2, \omega_3\})^c = \{\omega_1, \omega_2\}$$

$$K_1 (K_2 \{\omega_2, \omega_3\})^c = K_1 \{\omega_1, \omega_2\} = \{\omega_1\}$$

Only at ω_1 P1 knows that P2 doesn't know the game

$$\hookrightarrow K_2 \{\omega_2, \omega_3\} = \{\omega_3\}$$

$$K_1 (K_2 \{\omega_2, \omega_3\}) = K_1 \{\omega_3\} = \emptyset$$

At ω_3 P1 doesn't know if P2 knows the game or not.

Suppose that $T_1 = \{a, b\}$, $T_2 = \{a', b'\}$. Suppose the type maps are given by
 $\tau_1(\omega_1) = a$, $\tau_1(\omega_2) = \tau_1(\omega_3) = b$, and $\tau_2(\omega_1) = \tau_2(\omega_2) = a'$, $\tau_2(\omega_3) = b'$

$$\Theta = \{\theta_1, \theta_2\}$$

$$\omega_1 \leftrightarrow \theta_1$$

$$\omega_2, \omega_3 \leftrightarrow \theta_2$$

		Column	
		L	R
Row	U	2, 2	0, 0
	D	3, 0	1, 1

ω_1 : prob. $9/13$

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_2 : prob. $3/13$

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_3 : prob. $1/13$

$$P1 : S_1(a), S_1(b)$$

$$P2 : S_2(a'), S_2(b')$$

$S_1^*(a) = D$ because it is a dominant strategy for P1

Now consider P2 of type a' : $\Pr(\omega_1 | a') = \frac{9/13}{9/13 + 3/13} = 3/4$
 $\Pr(\omega_2 | a') = 1/4$

Playing L : $\frac{3}{4} \cdot (0) + \frac{1}{4} \cdot (2) = \frac{1}{2}$ } $\Rightarrow S_2^*(a') = R$
 gets at most

Playing R : $\frac{3}{4} \cdot (1) + \frac{1}{4} \cdot (0) = \frac{3}{4}$ }
 gets at least

Suppose that $T_1 = \{a, b\}$, $T_2 = \{b', b''\}$. Suppose the type maps are given by
 $\tau_1(\omega_1) = a$, $\tau_1(\omega_2) = \tau_1(\omega_3) = b$, and $\tau_2(\omega_1) = \tau_2(\omega_2) = a'$, $\tau_2(\omega_3) = b'$

$\Theta = \{\theta_1, \theta_2\}$

		Column	
		L	R
Row	U	2, 2	0, 0
	D	3, 0	1, 1

ω_1 : prob. 9/13

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_2 : prob. 3/13

		Column	
		L	R
Row	U	2, 2	0, 0
	D	0, 0	1, 1

ω_3 : prob. 1/13

$$s_1^+(a) = D, \quad s_2^+(a') = R$$

P1 of type b: $\Pr(\omega_2 | b) = \frac{\frac{3}{13}}{\frac{3}{13} + \frac{1}{13}} = \frac{3}{4}$
 $\Pr(\omega_3 | b) = \frac{1}{4}$

U gets at most $\frac{3}{4}(0) + \frac{1}{4}(2) = \frac{1}{2}$
D gets at least $\frac{3}{4}(1) + \frac{1}{4}(0) = \frac{3}{4}$

} $s_1^+(b) = D$

Now consider P2 of type b' : $s_2^+(b') = R$

(U,L) is a PE outcome that is not at, even when both of them know the game.

Example: behavioral strategies

There are two players. Suppose that $T_1 = \{a, b\}$, $T_2 = \{c\}$, $\tau_1(t) = t_1$. The common prior is $\mu(a, c) = \mu(b, c) = 1/2$. The state games corresponding to each type vector are:

$$\Omega = T_1 \times T_2 = \{(a, c), (b, c)\}$$

$$\tau_1(t) = t_1$$

$$\tau_2(t) = t_2$$

		Column	
		L q	R $1-q$
Row	T x	1, 0	0, 2
	B $1-x$	0, 3	1, 0

State game for $t = (a, c)$

		Column	
		L q	R $1-q$
Row	T y	0, 2	1, 1
	B $1-y$	1, 0	0, 2

State game for $t = (b, c)$

$$q = \Pr(\text{P2 of type } c, \text{ plays } L)$$

$$x = \Pr(\text{P1 of type } a, \text{ plays } T)$$

$$y = \Pr(\text{P1 of type } b, \text{ plays } T)$$

There are two players. Suppose that $T_1 = \{a, b\}$, $T_2 = \{c\}$, $\tau_1(t) = t_1$. The common prior is $\mu(a, c) = \mu(b, c) = 1/2$. The state games corresponding to each type vector are:

$$\Omega = T_1 \times T_2 = \{(a, c), (b, c)\}$$

		Column	
		L q	R $1-q$
Row $x \in T_1$ $x \in B$	T	1, 0	0, 2
	B	0, 3	1, 0

State game for $t = (a, c)$

		Column	
		L q	R $1-q$
Row $y \in T_2$ $y \in B$	T	0, 2	1, 1
	B	1, 0	0, 2

State game for $t = (b, c)$

$$q = \Pr(\text{P2 of type } c, \text{ plays } L)$$

$$x = \Pr(\text{P1 of type } a, \text{ plays } T)$$

$$y = \Pr(\text{P1 of type } b, \text{ plays } T)$$

Player 1

$$x = 1 \text{ (T purely)} \quad 1 \cdot q + 0 \cdot (1-q) > 0 \cdot q + 1 \cdot (1-q)$$

$$y = 1 \text{ (T purely)} \quad 0 \cdot q + 1 \cdot (1-q) > 1 \cdot q + 0 \cdot (1-q)$$

Player 2

$$L: \frac{1}{2} [x \cdot 0 + (1-x) \cdot 3] + \frac{1}{2} [y \cdot 2 + (1-y) \cdot 0]$$

$$R: \frac{1}{2} [x \cdot 2 + (1-x) \cdot 0] + \frac{1}{2} [y \cdot 1 + (1-y) \cdot 2]$$

$$BR_1(q; t_1) = \begin{cases} x=1 & \text{if } q > 1/2 \\ x \in [0, 1] & \text{if } q = 1/2 \\ x=0 & \text{if } q < 1/2 \end{cases} \quad t_1 = a$$

$$\begin{cases} y=1 & \text{if } q < 1/2 \\ y \in [0, 1] & \text{if } q = 1/2 \\ y=0 & \text{if } q > 1/2 \end{cases} \quad t_1 = b$$

$$BR_2(x, y) = \begin{cases} q=1 & \text{if } 1 > 5x-3y \\ q \in [0, 1] & \text{if } 1 = 5x-3y \\ q=0 & \text{if } 1 < 5x-3y \end{cases}$$

There are two players. Suppose that $T_1 = \{a, b\}$, $T_2 = \{c\}$, $\tau_1(t) = t_1$. The common prior is $\mu(a, c) = \mu(b, c) = 1/2$. The state games corresponding to each type vector are:

$\Omega = T_1 \times T_2 = \{(a, c), (b, c)\}$

Column

	L	R
Row		
$y=T$	1, 0	0, 2
$y=B$	0, 3	1, 0

State game for $t = (a, c)$

Column

	L	R
Row		
$y=T$	0, 2	1, 1
$y=B$	1, 0	0, 2

State game for $t = (b, c)$

$q = P_i(P_i \text{ of type } c, \text{ plays } L)$
 $x = P_i(P_i \text{ of type } a, \text{ plays } T)$
 $y = P_i(P_i \text{ of type } b, \text{ plays } T)$

$$BR_1(q; t_1) = \begin{cases} x=1 & \text{if } q > 1/2 \\ x \in [0, 1] & \text{if } q = 1/2 \\ x=0 & \text{if } q < 1/2 \end{cases} \quad t_1 = a$$

$$\begin{cases} y=1 & \text{if } q < 1/2 \\ y \in [0, 1] & \text{if } q = 1/2 \\ y=0 & \text{if } q > 1/2 \end{cases} \quad t_1 = b$$

$$BR_2(x, y) = \begin{cases} q=1 & \text{if } 1 > 5x-3y \\ q \in [0, 1] & \text{if } 1 = 5x-3y \\ q=0 & \text{if } 1 < 5x-3y \end{cases}$$

If $q=1 \Rightarrow x=1, y=0 \Rightarrow q=0$!

If $q=0 \Rightarrow x=0, y=1 \Rightarrow q=1$!

No pure strategy eq.

$$q < 1/2 \Rightarrow x=0, y=1 \Rightarrow q=1 !$$

$$q > 1/2 \Rightarrow x=1, y=0 \Rightarrow q=0 !$$

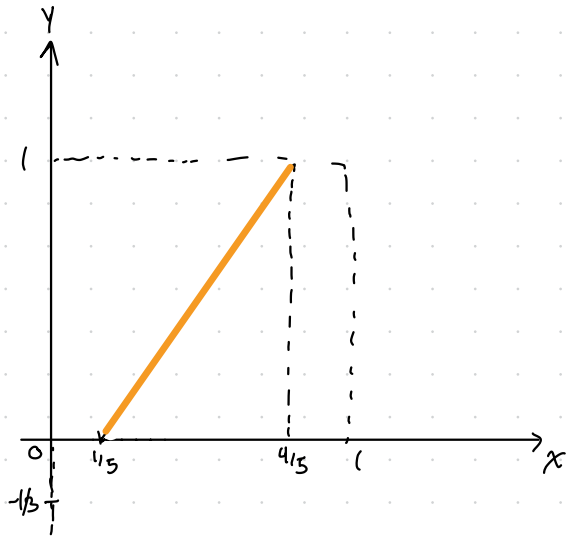
If $q = 1/2; (x, y) \in [0, 1]^2$

as long as they satisfy $1 = 5x - 3y$

this is an eq.

The set of BNE is such that

$$\{(q, x, y) : q = 1/2, x = \frac{1+3y}{5}, y \in [0, 1]\}$$



$$1 = 5x - 3y$$

$$y = \frac{5x - 1}{3}$$

$$y = -\frac{1}{3} + \frac{5}{3}x$$

Example: Cournot's Duopoly with cost uncertainty

Two firms compete in selling a good: one firm does not know the other firm's cost function. Assume that both firms can produce the good at a constant unit cost. Assume also that they both know that firm 1's unit cost is c , but only firm 2 knows its own unit cost; firm 1 believes that firm 2's cost is c_L with probability θ and c_H with probability $1 - \theta$, where $0 < \theta < 1$ and $c_L < c_H$. Assume also that the inverse demand function is $P(Q)$ with $P(Q) > 0$ if $Q \in [0, \bar{Q})$ and 0 otherwise. Assume also that $P(Q)$ is differentiable.

$\Omega = \{L, H\}$ $T_1 = \{c\}$ Strategies: q_1

$T_1(L) = T_1(H)$ $T_2 = \{c_L, c_H\}$ $q_2(c_L), q_2(c_H)$

$T_2(L) \neq T_2(H)$

Payoffs: $P_1: \theta [P(q_1 + q_{2,L}) q_1 - c q_1] + (1-\theta) [P(q_1 + q_{2,H}) q_1 - c q_1]$

$P_2: P(q_1 + q_{2,i}) q_{2,i} - c_i q_{2,i} \quad i = L, H$